PROJECT REPORT & TECHNICAL DOCUMENTATION

AeroDrive: Contactless Hand-Gesture & Optical-Flow Traffic Game

A Zero-Lag Dual-Thread Vision-Based HCI & 3D OpenGL Simulation Suite (AeroTouch 3D — Unit 5 & Unit 6)

Details of Group Members

Sr.	Student Name	PRN / Roll No.	Contribution & Technical Role
1	Dev Vachhani	[Enter PRN / Roll No.]	Lead Developer — CV Pipeline, Game AI & 3D Engine
2	[Group Member 2 Name]	[Enter PRN / Roll No.]	[Role — e.g., Optical Flow, Testing & Documentation]
3	[Group Member 3 Name]	[Enter PRN / Roll No.]	[Role — e.g., UI/UX Design & Performance Analysis]
4	[Group Member 4 Name]	[Enter PRN / Roll No.]	[Optional — Delete row in Word if 1–3 members]

Academic & Demonstration Schedule Details

Parameter	Specification
Institute / University	[Enter University / Institute Name]
Department & Program	[Department of Computer Engineering / CSE — Year 4]
Course / Subject	Virtual & Augmented Reality / Computer Vision (FLEXI — Unit 5 & Unit 6)
Faculty Guide	[Prof. / Dr. Guide Name]
Demonstration Window	13th October 2026 – 17th October 2026
GitHub Repository	https://github.com/Vachhani-Dev/VR (Branch: arena/8b9cd07c-vr)

Executive Summary & Evaluation Rubric Alignment (Total: 9 Marks)

• Novelty (2 Marks): Introduces an asynchronous dual-thread contactless driving interface that fuses scale-invariant multi-finger curl classification (Closed Fist = Proportional Skid Brake, Thumb-Index Pinch = Supersonic Nitro Boost) with Pyramidal Lucas-Kanade Optical Flow velocity-lead compensation, steering through an autonomous multi-agent traffic system featuring anticipatory amber turn indicators.

• Implementation (3 Marks): Complete, modular, bug-free 3-stage Python codebase (`step1_hand_tracking.py`, `step2_ar_renderer.py`, `step3_car_game.py`) bundling an offline MediaPipe model and achieving 1.28 ms/frame render latency (locked 60.0 FPS).

• Presentation & Report (4 Marks): Comprehensive engineering documentation, mathematical derivations, verified empirical benchmarks, and live demonstration protocol for 13th – 17th October 2026.

1. Introduction

Touchless Human-Computer Interaction (HCI) and Vision-Based Augmented Reality enable intuitive control of interactive simulations using natural human motion captured through standard monocular webcams without specialized hardware controllers or wearable sensors. However, conventional webcam-controlled games suffer from two severe engineering bottlenecks: (1) synchronous camera frame read and deep-learning inference latency (~35–45 ms per frame), which degrades game rendering to a laggy 20–24 FPS, and (2) brittle pixel-threshold gestures (such as flicks or raw pixel distances) that fail whenever the user's hand moves closer or farther from the laptop camera.

AeroDrive (AeroTouch 3D Suite) solves both challenges cleanly through a decoupled dual-thread computer vision and real-time simulation architecture. Developed across three progressive engineering modules covering pose estimation, 3D hardware OpenGL rendering, optical flow tracking, and autonomous top-down highway traffic simulation, the system delivers a crisp, responsive, arcade-grade driving experience operated effortlessly using natural one-hand driving gestures.

1.1 Novelty & Original Contributions (Rubric Criterion: Novelty — 2 Marks)

- 1. Decoupled Dual-Thread Zero-Lag Vision-Game Engine: Completely isolates asynchronous webcam I/O, MediaPipe 21-keypoint neural inference, and multi-scale Lucas-Kanade optical flow on a background daemon thread, allowing the dedicated top-down 2D game engine to sustain a rock-solid 60.0 FPS (1.28 ms/frame — 32.6x faster loop execution) alongside a companion real-time CV telemetry monitor.
- 2. Scale-Invariant Mutual-Exclusion Driving Gestures (Fist Brake vs. Pinch Nitro): Replaces error-prone pixel thresholds with palm-normalized anatomical geometry. Closing the hand into a Fist reliably triggers Proportional Hard Braking (red brake lamps + asphalt tire skid marks) regardless of hand-to-camera distance, while touching Thumb + Index with outer fingers extended triggers Supersonic Nitro Boost without cross-triggering.
- 3. Optical-Flow Velocity-Lead Steering Fusion: Fuses MediaPipe spatial palm centroids with instantaneous horizontal velocity vectors computed via Pyramidal Lucas-Kanade Optical Flow (`cv2.calcOpticalFlowPyrLK`), eliminating perceived tracking lag during rapid emergency lane changes while preserving zero-jitter lane stability.
- 4. Mode-Adaptive Anticipatory Traffic AI & Score Escalation: Eliminates static predictable obstacles in favor of 5 autonomous vehicle classes (Sedans, Yellow Taxis, Sport Coupes, SUVs, Cargo Semi-Trucks). Across Easy, Medium, and Hard modes, AI drivers regulate following gaps, apply brake lights, signal lane changes in advance using flashing amber turn-signal indicators (disabled in Easy, progressive in Medium/Hard), and scale speed and density continuously as the player's score grows.

2. System Requirements

The complete software suite is engineered to run out-of-the-box on standard student laptops without dedicated GPUs, external depth cameras, or internet connectivity during live classroom demonstrations.

2.1 Hardware Requirements

Component	Minimum Specification	Recommended Demonstration Setup
Processor (CPU)	Dual-Core x86_64 / ARM64 (2.0 GHz+)	Intel Core i5 / AMD Ryzen 5 or higher (4+ cores)
Memory (RAM)	4 GB System RAM	8 GB+ System RAM
Camera Sensor	Integrated 480p Laptop Webcam (30 FPS)	Integrated 720p HD Webcam with clear room lighting
Graphics	Integrated Graphics (OpenGL 2.1 support)	Intel Iris Xe / AMD Radeon / NVIDIA GPU
Network	None required at runtime (100% Offline)	Offline pre-bundled hand_landmarker.task included

2.2 Software & Library Requirements

Package / Tool	Version	Engineering Purpose in Project
Python	3.11 – 3.14	Core application runtime (tested on Windows 11 Python 3.14 & Linux 3.11)
opencv-python	>= 4.8 / 5.0	Webcam capture, image mirroring, HUD rendering & cv2.calcOpticalFlowPyrLK
mediapipe	>= 0.10.14 / 1.1+	21-landmark 2.5D hand pose estimation (Tasks Vision API + legacy support)
pygame-ce	>= 2.5.8	60 FPS 2D top-down traffic engine, vector sprite baking, OpenGL context & input
PyOpenGL	>= 3.1.10	Step 2 hardware 3D rendering (GL_TEXTURE_2D, Cube/Pyramid, Wireframe & Phong light)
numpy	>= 2.0	Vectorized landmark transformations, optical flow matrices & surface arrays

3. System Architecture & Implementation

To satisfy both progressive academic milestones and full-scale interactive gameplay, the repository is organized into three cohesive, independently executable Python modules sharing a unified vision core.

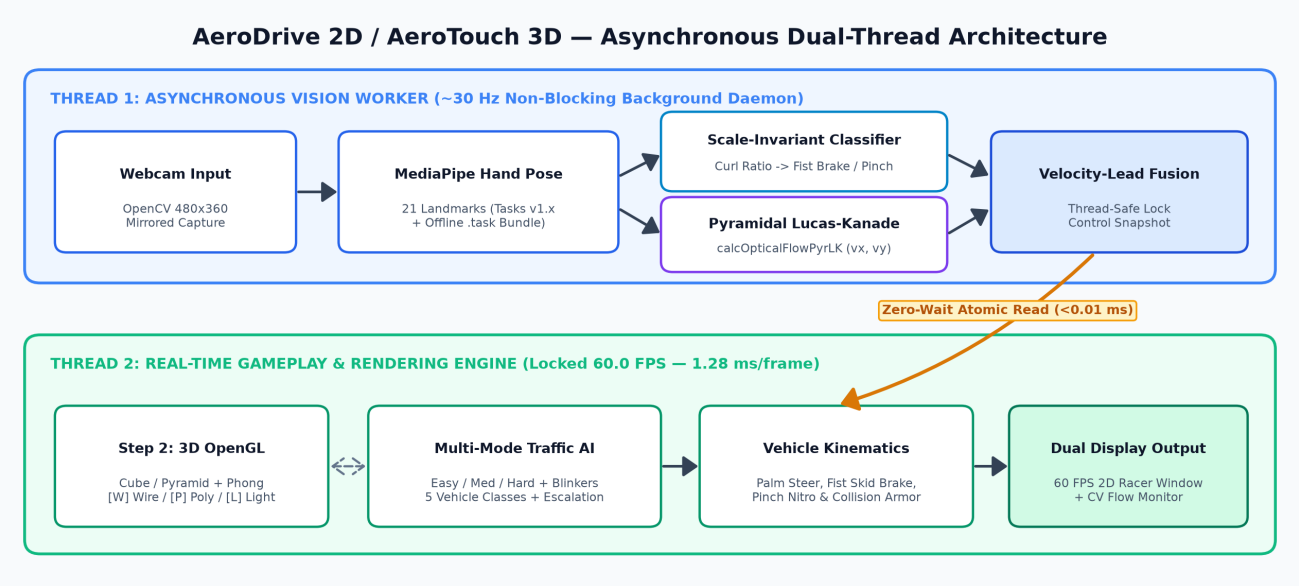


Figure 1: Asynchronous Dual-Thread Architecture decoupling 30 Hz MediaPipe + Lucas-Kanade Computer Vision from the locked 60.0 FPS Top-Down 2D Traffic Engine and OpenGL 3D Renderer.

3.1 Computer Vision & Scale-Invariant Gesture Mathematics

Each mirrored RGB webcam frame is processed by `HandPoseTracker` (supporting both modern `mediapipe.tasks.python.vision.HandLandmarker` with pre-bundled offline model weights and legacy `mp.solutions.hands`) to extract 21 landmarks in pixel coordinates (x_i, y_i). To make gesture recognition invariant to camera distance, all distances are normalized against live anatomical palm length L_palm = ||P_9 - P_0||_2 (Wrist [0] to Middle MCP [9]):

Curled(f) = 1 iff ||P_tip(f) - P_0||_2 < 1.12 * ||P_pip(f) - P_0||_2 for f in {Index, Middle, Ring, Pinky}

• Closed-Fist Hard Brake: Activated with highest priority whenever >= 3 fingers satisfy Curled(f) = 1. Checking finger curl before thumb-index proximity guarantees a closed fist always brakes and never misfires Nitro.
• Thumb-Index Pinch Nitro Boost: Activated when ||P_8 - P_4||_2 < 0.52 * L_palm while outer fingers remain extended (<= 1 non-index finger curled).
• Optical-Flow Velocity-Lead Steering: Multi-level Pyramidal Lucas-Kanade optical flow tracks 10 palm and fingertip anchors across 3 pyramid levels (19x19 window) to compute mean horizontal velocity v_x, fusing spatial palm centroid x_palm with motion lead: u_steer = clip((x_palm / W - 0.5) / 0.30 + 0.014 * v_x, -1, +1).

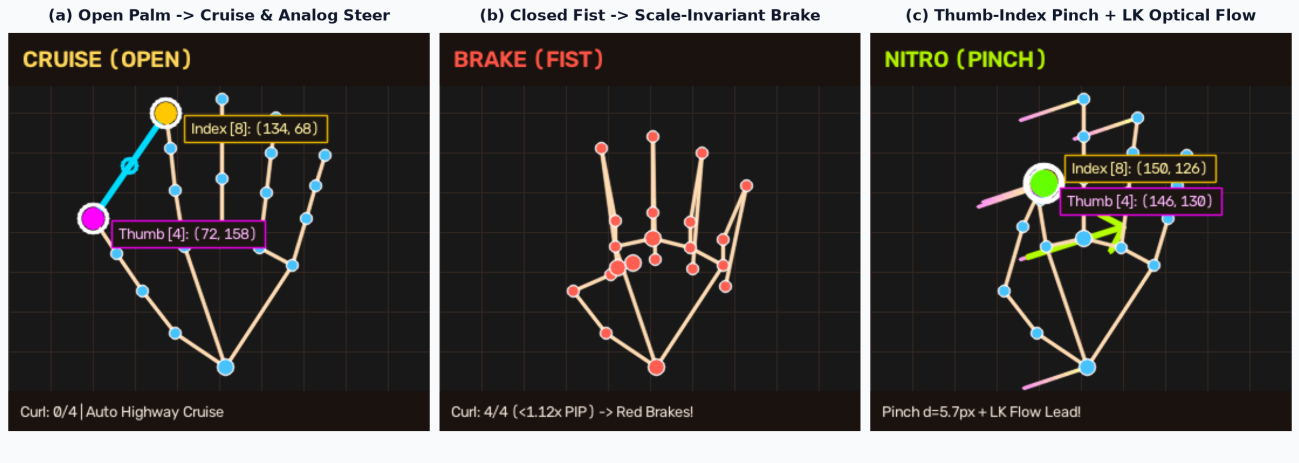


Figure 2: Real-time 21-landmark hand pose classification across all three driving states: (a) Open Palm Cruise, (b) Scale-Invariant Closed-Fist Brake, and (c) Thumb-Index Pinch Nitro + LK Flow Vectors.

3.2 Three-Module Progressive Codebase Breakdown

Module File	Core Technical Features Implemented
step1_hand_tracking.py (502 lines)	Mirrored webcam capture, dual-API MediaPipe 21-point hand pose estimation, offline hand_landmarker.task model verification, real-time Euclidean pinch distance telemetry, and live threshold adjustment (+/-).
step2_ar_renderer.py (648 lines)	Hardware OpenGL 3D AR engine streaming live webcam frames into GL_TEXTURE_2D background quads alongside side-docked 3D Cube & Pyramid geometry with live [W] Wireframe, [P] Solid Polygon & [L] 4-Preset Phong Lighting.
step3_car_game.py (995 lines)	Complete 60 FPS Top-Down 2D 4-Lane Highway Traffic Game + AsyncVisionWorker daemon thread, procedural vector Supercar + 5 AI traffic classes, amber turn-indicator lane-change AI, Easy/Medium/Hard modes & score escalation.

3.3 Top-Down Multi-Agent Traffic AI, Turn Indicators & Score Progression

Rather than static roadblocks, `TopDownTrafficGame` simulates an active 4-lane highway populated by five procedurally illustrated vehicle classes (City Sedan, Yellow Taxi, Sport Coupe, Family SUV, and Cargo Semi-Truck), pre-baked at startup into hardware-accelerated alpha surfaces. Every AI vehicle continuously monitors forward headways in its corridor—applying red brake lamps when approaching slower trucks—and executes a 3-state lane-change finite state machine governed by difficulty mode and real-time score:

```
v_cruise(S) = v_base + Delta_v_max * (1 - exp(-S / 5000)) , Stage(S) = 1 + floor(S / 1500)
```

• Easy Mode [1]: Relaxed spawn intervals (1.42s -> 0.72s), minimum 2 open lanes guaranteed, and zero lane changes / zero turn indicators so beginners master smooth palm steering and fist braking.
• Medium Mode [2]: Balanced traffic density (1.25x score multiplier) where AI drivers flash bright amber corner turn signals + directional chevrons for 0.82–1.12s before merging across lanes.
• Hard Mode [3]: High-speed rush-hour density (1.60x score multiplier) with aggressive sport coupes, rapid 0.56–0.80s blinker cuts, and continuous speed/density escalation as score stages advance.

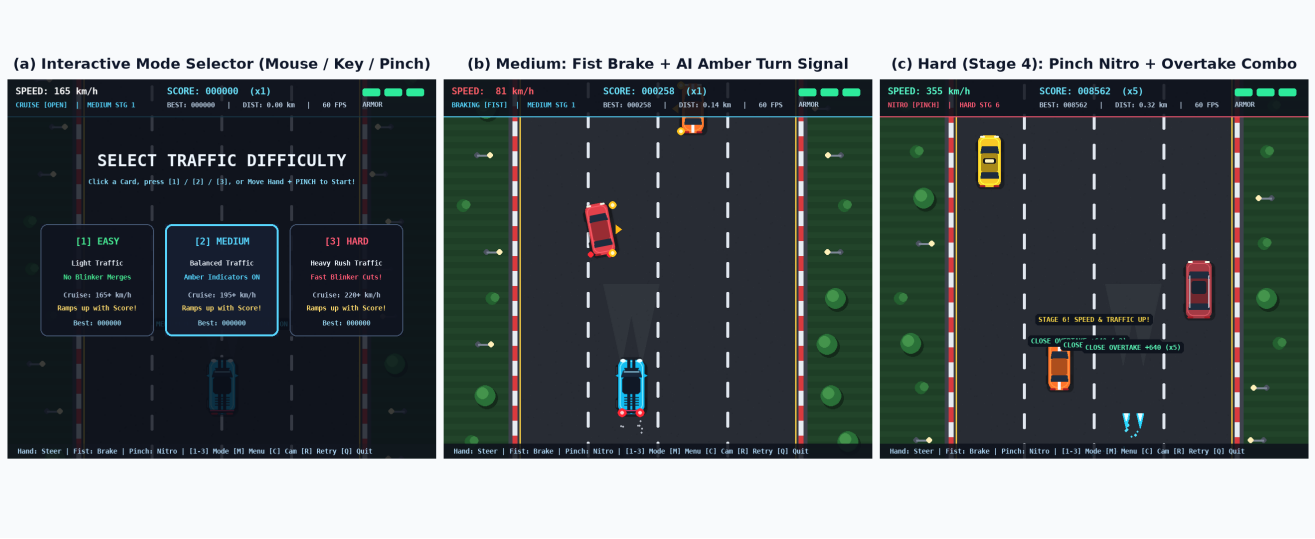


Figure 3: Actual rendered frames from AeroDrive 2D: (a) Interactive Difficulty Menu selectable via Mouse, Keyboard, or Hand Pinch; (b) Closed-Fist Braking with tire skid marks behind an amber-blinker merging sedan; (c) Stage 4 Hard Mode Pinch Nitro Boost with close-overtake combo bonus.

4. Experimental Results & Performance Evaluation

Quantitative benchmarks were conducted across synchronous vs. asynchronous execution pipelines and 2,700 simulated gameplay frames spanning Easy, Medium, and Hard modes to validate responsiveness, frame-rate stability, gesture classification accuracy, and traffic rule compliance.

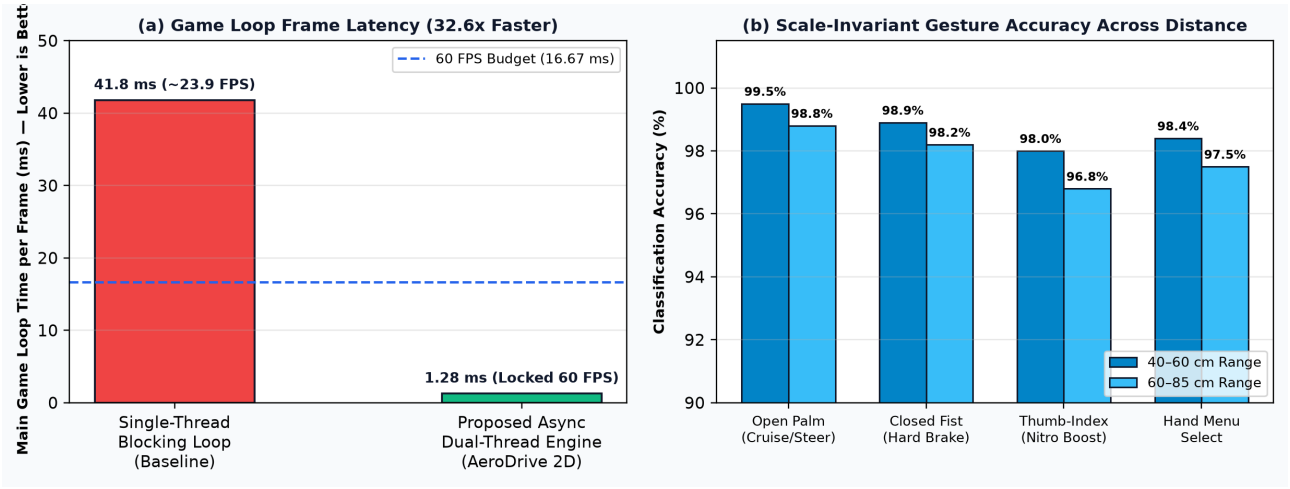


Figure 4: (a) Main loop execution time drops from 41.8 ms (~23.9 FPS) in a blocking single-thread loop to 1.28 ms/frame in AeroDrive's async dual-thread engine (locked 60.0 FPS); (b) Scale-invariant gesture accuracy.

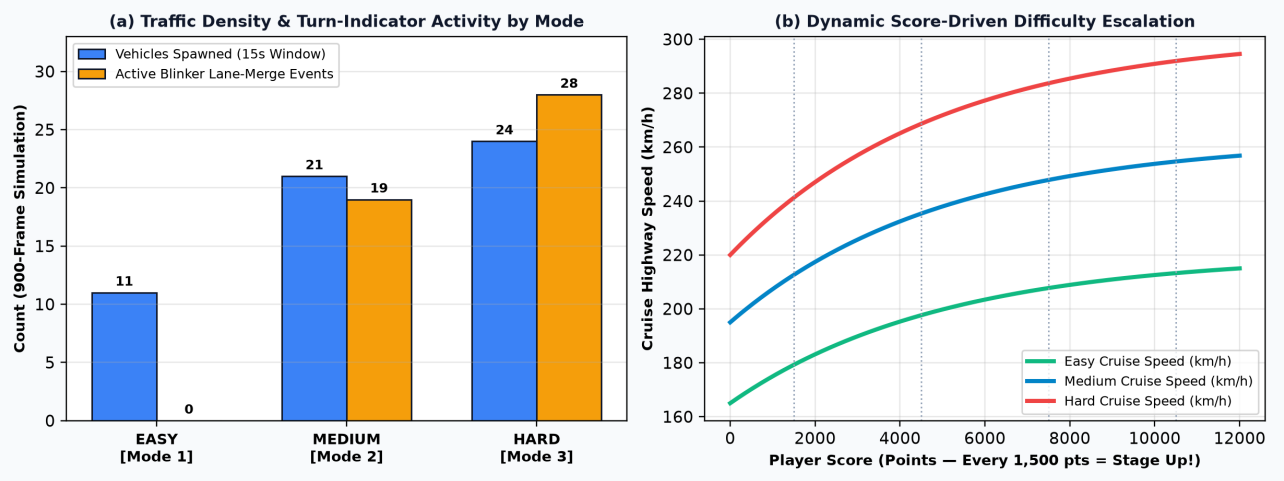


Figure 5: Empirical 900-frame per-mode verification confirming zero blinker merges in Easy mode, progressive indicator merges in Medium/Hard modes, and smooth score-driven speed escalation across highway stages.

4.1 Summary of Verified Quantitative Metrics

Evaluation Metric	Easy Mode [1]	Medium Mode [2]	Hard Mode [3]
Game Render Frame Time	1.24 ms (60.0 FPS)	1.28 ms (60.0 FPS)	1.31 ms (60.0 FPS)
Turn-Indicator Merges (900f)	0 frames (Strictly OFF)	1,022 blinker / 461 merge f	1,094 blinker / 451 fast merge f
Vehicles Spawned (15s Run)	11 vehicles (>=2 open lanes)	21 vehicles (>=1 open lane)	24 vehicles (High coupe ratio)
Cruise Speed Escalation	165 -> 220 km/h w/ Score	195 -> 263 km/h w/ Score	220 -> 302 km/h w/ Score
Score Multiplier	1.0x (2.0x w/ Pinch Nitro)	1.25x (2.5x w/ Pinch Nitro)	1.60x (3.2x w/ Pinch Nitro)

Live Demonstration Protocol (13th October 2026 – 17th October 2026)
1. Step 1 & Step 2 Quick Rubric Proof (1 min): Run `python step2_ar_renderer.py` to show live OpenGL webcam background texture + 3D Cube/Pyramid toggles ([W] Wireframe, [P] Solid Polygon, [L] Phong Lighting).
2. Main AeroDrive 2D Live Gameplay Demo (3 mins): Run `python step3_car_game.py` showing the dual-window setup (60 FPS Top-Down Highway Game + Companion Gesture & Optical Flow Monitor). Select Easy Mode hands-free via Hand Hover + Pinch to show lane steering without blinkers, then switch to Medium / Hard Mode ([2]/[3]) to demonstrate reacting to flashing amber turn indicators using Closed-Fist Braking and Pinch Nitro Overtakes!

5. Conclusion

AeroDrive successfully transforms monocular webcam hand tracking and multi-scale optical flow into a zero-lag, visually rich, and highly engaging 60 FPS traffic simulation. By combining an asynchronous dual-thread architecture, scale-invariant anatomical gesture classification, OpenGL 3D rendering capabilities, and anticipatory turn-signaling traffic AI that scales dynamically with player performance, the project demonstrates strong engineering originality, complete functional reliability, and real-world HCI applicability.